

GEORGE JOBI PERANGATTU

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PROFESSIONAL SUMMARY

C++ Autonomous Systems Software Engineer pursuing an M.S. in Robotics and Autonomous Systems (AI) at Arizona State University, with a B.S. in Computer Science. Hands-on experience implementing autonomy software in modern C/C++ and Python across perception, localization, state estimation, motion planning, controls, simulation, sensor integration, testing, and physical robot validation. Strong Linux and software-engineering foundation with ROS/ROS 2 concepts, Gazebo, CARLA, algorithms, data structures, debugging, APIs, CI/CD, Docker, documentation, and production code.

EDUCATION

M.S. Robotics and Autonomous Systems (AI Concentration) Graduating Jul 2026

Arizona State University, Tempe, AZ

B.S. Computer Science Graduated May 2025

Arizona State University, Tempe, AZ

TECHNICAL SKILLS

Role Emphasis: Modern C++, C++11 and Above, Autonomous Systems, Onboard Application Software, Physical AI, Robotics, Interface Contracts, Execution Models, Cross-Stack Integration, Embedded Software Concepts, Real-Time Systems Concepts, Python Tooling, Testing, Debugging

Localization & SLAM: Robot Localization, Localization Algorithms, SLAM Fundamentals, State Estimation, Kalman Filter Fundamentals, Particle Filter Fundamentals, Sensor Fusion, Camera-LiDAR Fusion, Pose Estimation, Depth Estimation, Mapping, Navigation, Positioning

Robotic Sensor Systems: LiDAR, Stereo Cameras, RGB Cameras, RGB-D Cameras, Depth Sensors, Point Cloud Processing, Object Detection, Segmentation, BEV Perception, 3D Perception, Computer Vision, OpenCV, Sensor Integration

Autonomous Robotics Software: ROS/ROS 2, Gazebo Simulation, CARLA, URDF, Motion Planning (RRT), Path Planning, Trajectory Optimization, Collision Avoidance, Kinematics, PID Control, Closed-Loop Control, Autonomous Systems

Languages & Platforms: C/C++ (Modern), Python, C, Java, JavaScript, TypeScript, SQL, Linux-Based Platforms, Linux/Ubuntu, Embedded Platforms Fundamentals

ML & Deep Learning: PyTorch, TensorFlow, Transformers, CNNs, Multi-Task Learning, Mixed-Precision Training, Machine Learning, Data-Driven Analysis, NumPy, Pandas

Software Engineering: Git, CMake, CI/CD, Docker, Code Review, Agile, Testing, Validation, Production Code, Documentation, Problem-Solving, GDB, Algorithms, Data Structures

LOCALIZATION & PERCEPTION PROJECT EXPERIENCE

Efficient TransFuser: Camera-LiDAR Sensor Fusion for Localization & Perception Spring 2026

Arizona State University - CSE 598 Research Project

- Emphasized autonomous-driving perception and localization software, camera-LiDAR integration, performance optimization, and robust testing.
- Developed and optimized robot localization and perception algorithms by designing a camera-LiDAR sensor fusion pipeline, replacing TransFuser's backbone with EfficientNetV2-S to reduce parameters by 70% (168M to 50M) while improving held-out loss by 8.7% and working with LiDAR, stereo cameras, and RGB cameras for autonomous robotic systems.
- Built multi-scale transformer fusion for RGB image and LiDAR point cloud processing, achieving 25.4% BEV segmentation improvement for object detection, pose estimation, depth estimation, and localization in the CARLA autonomous driving simulator.
- Designed data loading and training pipelines for multi-task robotics perception across navigation, depth estimation, semantic segmentation, detection, and BEV mapping, using synthetic data generation and data-driven analysis for model evaluation.
- Ran model evaluations with cosine LR scheduling, gradient clipping, and Kendall uncertainty weighting, writing reusable code and organizing results across multi-task experiments for localization, perception, and navigation outputs.
- Diagnosed an AMP instability in CenterNet's focal loss under FP16 GPU training, documenting, validating, and maintaining high-quality production code with rigorous testing and problem-solving methodology.

PROJECT EXPERIENCE (CONTINUED)

Parrot MiniDrone - Real-Time Localization & Autonomous Control Spring 2026

Arizona State University - Robotics & Autonomous Systems Coursework

- Emphasized onboard physical-robot control, state estimation, sensor fusion, real-time behavior, safety checks, and hardware validation.
- Developed robot localization and navigation algorithms on a physical robotic platform, implementing closed-loop PID control (yaw, velocity, altitude) with sensor fusion of inertial and visual sensor data for real-time state estimation and positioning on Linux-based embedded hardware.
- Built robotics pipelines involving sensor fusion and model inference for automated test sequences, logging sensor data from robotic sensor systems and measuring localization, navigation, and control performance through iterative metrics.
- Engineered safety checks and failsafes ensuring reliable autonomous operation under real-time constraints, demonstrating strong problem-solving skills on real robotic hardware.

3D Motion Planning for Autonomous Navigation in Gazebo Fall 2025

Arizona State University - Robotics & Autonomous Systems Coursework

- Emphasized autonomy algorithms, C++-adjacent software architecture concepts, ROS/Gazebo interfaces, planning, and validation.
- Developed autonomous robotics software (3D RRT) in Python for collision-free trajectory computation, integrated with Gazebo simulation using ROS for testing, validation, and data-driven analysis of navigation and localization-relevant path planning performance.
- Applied path post-processing to optimize trajectory quality for downstream robotic control, designing experimental details and running evaluations to validate navigation algorithm performance across multiple environment configurations.

Spider CAD Robot - Multi-Legged Robot Design (In Progress) Spring 2026

Arizona State University - Robotics & Autonomous Systems Coursework

- Designing a multi-legged robot in CAD with constrained joints, validating kinematics and actuator placement; preparing a URDF-exportable model for navigation, localization, and controls experimentation in ROS/Gazebo simulation.

Factory Automation System for Semiconductor Process Simulation Starting Jun 2026

Arizona State University - Applied Project (Upcoming)

- Will develop an industrial automation system integrating robotic sensor systems, actuators, and basic PLC/control logic for a manufacturing process simulation, applying sensor integration and localization-relevant state tracking across hardware interfaces.

WORK EXPERIENCE

Backend Developer Intern - DigiClips Media Aug 2024 - May 2025

Tempe, AZ

- Developed ML models and data processing pipelines handling 10M+ daily queries using Python and PyTorch, reducing latency by 45% through data-driven analysis, testing, and performance optimization.
- Developed computer vision algorithms for image recognition with cross-functional teams, improving detection/perception accuracy by 30% through reusable code, model evaluation, validation, and iterative problem-solving.

Full Stack Developer Intern - Odoo May 2025 - Aug 2025

Dubai, UAE

- Refactored Python backend for production applications, contributing reusable infrastructure code and documenting, validating, and maintaining high-quality production code in collaboration with multidisciplinary engineering teams.

Full Stack Developer Intern - TicketDex Jan 2025 - May 2025

Tempe, AZ

- Delivered core backend services with testing infrastructure, achieving 30% latency reduction through algorithmic optimization, database refactoring, data-driven evaluation, and cross-functional collaboration.